

Industrial Internship Report on “Predictive Maintenance of Turbofan Engines using Machine Learning”

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on developing a predictive maintenance solution using Machine Learning. We had to finish the project including the report in 6 weeks' time.

My project was focused on analyzing time-series sensor data from the NASA CMAPSS dataset to predict the Remaining Useful Life (RUL) of turbofan jet engines. I developed an end-to-end Machine Learning pipeline utilizing the XGBoost algorithm and deployed it as an interactive web dashboard using Streamlit. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solutions for that. It was an overall great experience to have this internship.

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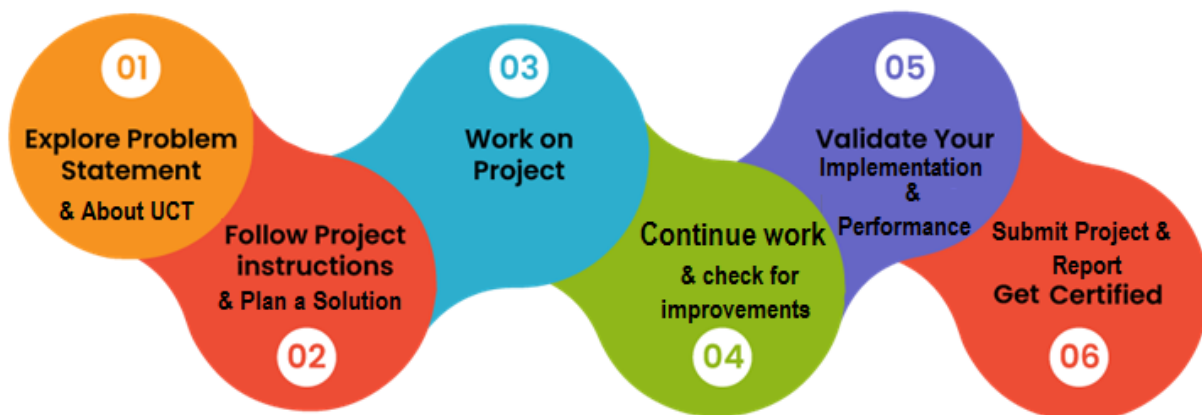
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1 Preface

During this 6-week internship, I transitioned from understanding basic data structures to developing industry-standard Machine Learning applications. As a Master of Computer Applications student, this practical exposure was crucial in bridging the gap between academic programming and real-world software engineering.

The project involved exploring the problem of unexpected industrial equipment failure. I developed a solution starting from a baseline Random Forest model, advanced to an XGBoost Regressor, and learned critical concepts like the "Curse of Dimensionality" during feature engineering. The opportunity provided by USC and UCT allowed me to experience the complete software lifecycle, culminating in deploying the model into a user-friendly Web App. I am deeply thankful to my mentors at upSkill Campus, UCT, and my university faculty for their continuous guidance. To my peers: always focus on understanding the data before jumping to complex algorithms.

How Program was planned



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoSaWAN), Java Full Stack, Python, Front end** etc.



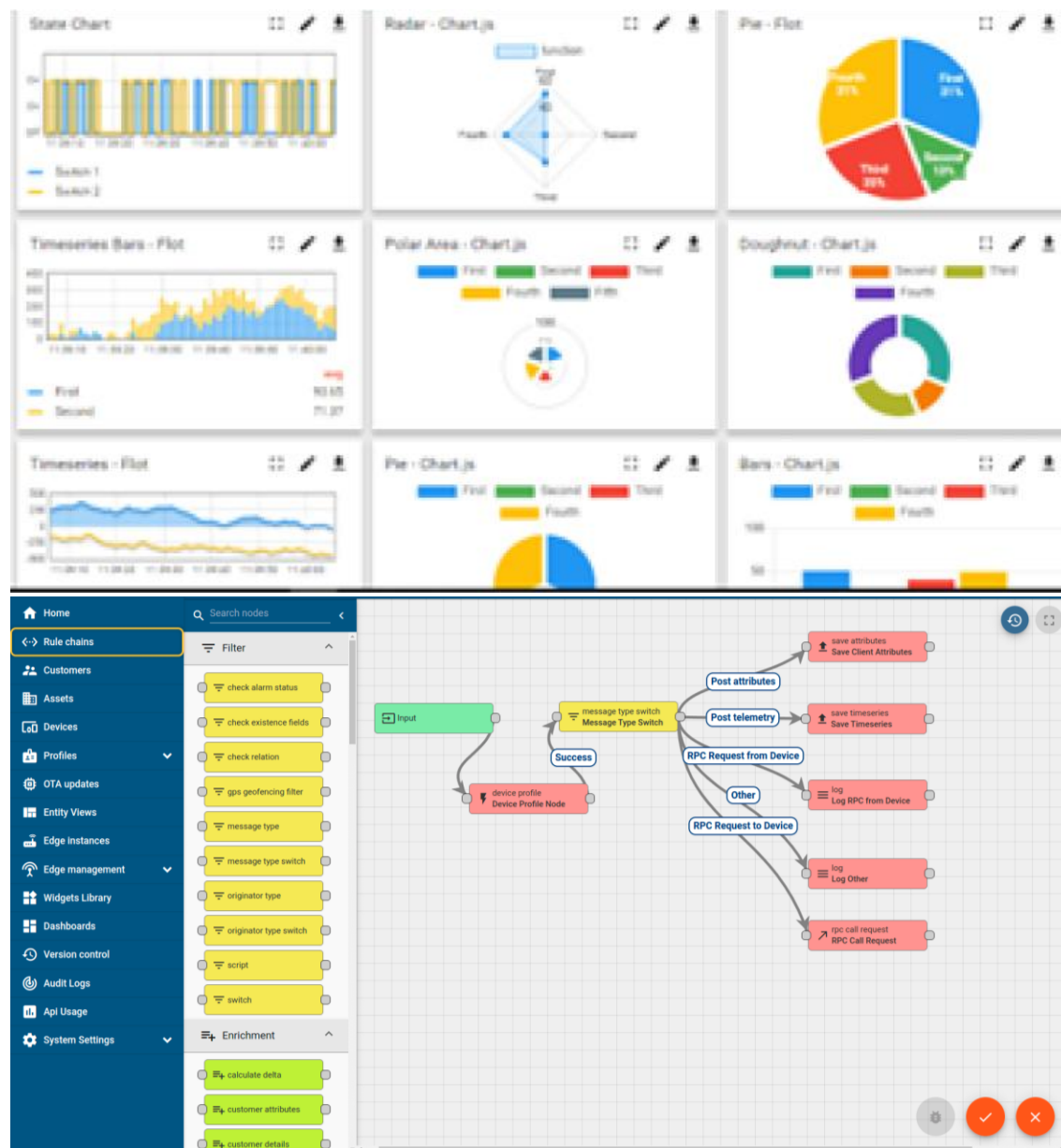
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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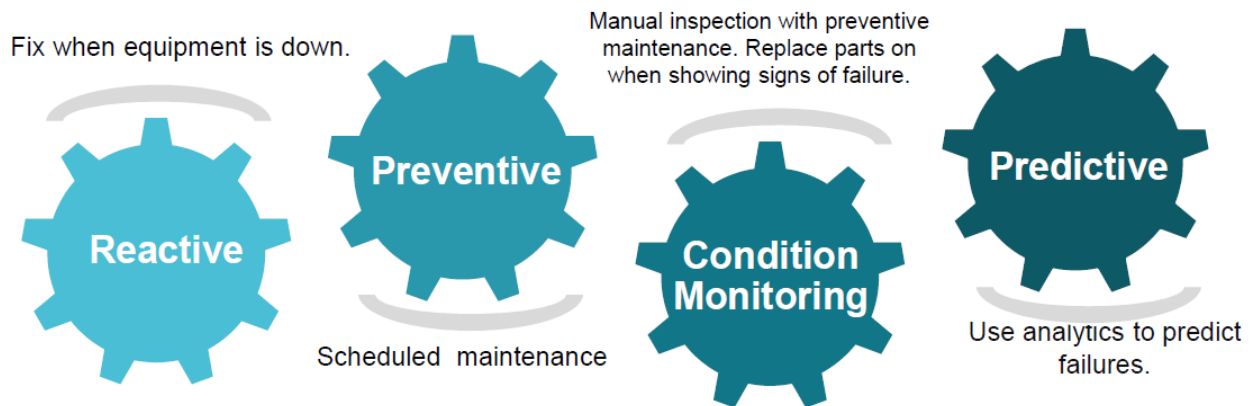


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

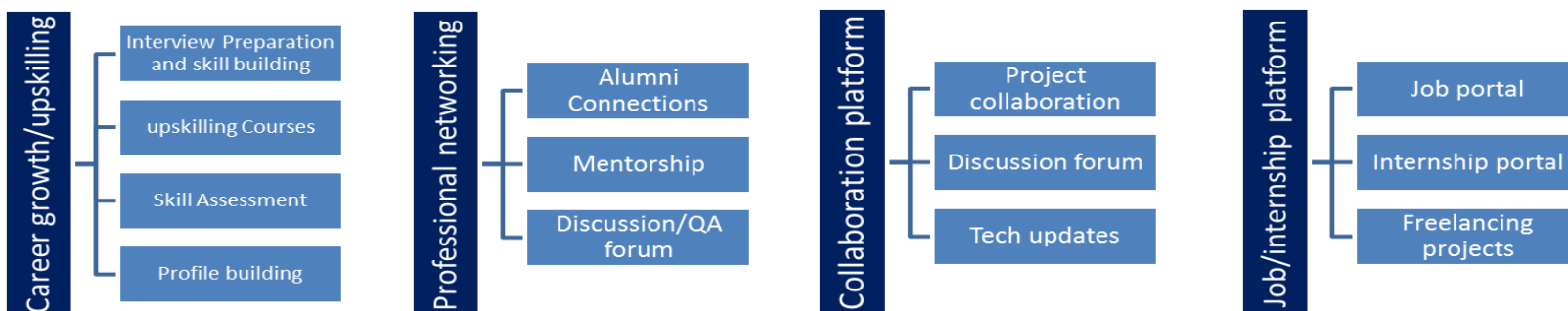
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] NASA CMAPSS Jet Engine Simulated Data (FD001 Dataset).
- [2] XGBoost Python Documentation (<https://xgboost.readthedocs.io/>)
- [3] Streamlit Application Framework Documentation (<https://docs.streamlit.io/>)

2.6 Glossary

Terms	Acronym
RUL	Remaining Useful Life
CMASS	Commercial Modular Aero-Propulsion System Simulation
XGBoost	Extreme Gradient Boosting
RMSE	Root Mean Square Error
UI	User Interface

3 Problem Statement

In the assigned problem statement

In the aviation and manufacturing industries, unexpected machine failure leads to massive financial losses and safety hazards. Currently, industries rely heavily on either "Reactive Maintenance" (fixing a machine after it breaks) or "Preventive Maintenance" (replacing parts on a fixed schedule, even if they are perfectly fine, leading to wasted resources).

The problem assigned was to implement "Predictive Maintenance". By analyzing multi-sensor time-series data (21 different sensors like temperature, pressure, and fan speed) across hundreds of operational cycles, the goal was to build a system that accurately predicts exactly *when* an engine will fail, thereby saving costs and preventing downtime.

4 Existing and Proposed solution

Existing Solutions & Limitations: Existing solutions often rely on single-sensor threshold alerts or basic statistical averages. Their limitation is that they cannot capture the complex, hidden patterns when multiple sensors degrade simultaneously. Furthermore, complex deep learning models are often deployed without a user interface, making them useless for floor managers.

Proposed Solution & Value Addition: I proposed and developed an XGBoost Regressor model. The key value addition is the "Piecewise Linear Degradation" technique, where RUL is clipped at 125 cycles because engines do not show wear-and-tear initially. Furthermore, I added massive value by packaging the AI model into a Streamlit Web Dashboard that provides real-time "Red Alerts" when the engine's RUL drops below 30 cycles.

4.1 Code submission (GitHub link) : [Click Here for Get Code File](#)

4.2 Report submission (GitHub link): [Click Here for Get Report File](#)

5 Proposed Design/ Model

The design flow of the system consists of 4 main stages:

1. **Data Preprocessing:** Loading `train_FD001.txt`, calculating actual RUL by grouping max cycles, and dropping constant operational settings (sensors 1, 5, 10, 16, 18, 19) to reduce noise.
2. **Model Training:** Training an XGBoost Regressor (`n_estimators=100`, `max_depth=5`) on the cleaned training data.
3. **Serialization:** Exporting the trained model locally as `turbofan_xgb_model.pkl` using Joblib to prevent version mismatch issues.
4. **Interface (Web App):** A Streamlit application `app.py` that accepts unseen test data, applies the exact same preprocessing, and visualizes the predicted RUL alongside sensor degradation trends.

5.1 High Level Diagram (if applicable)

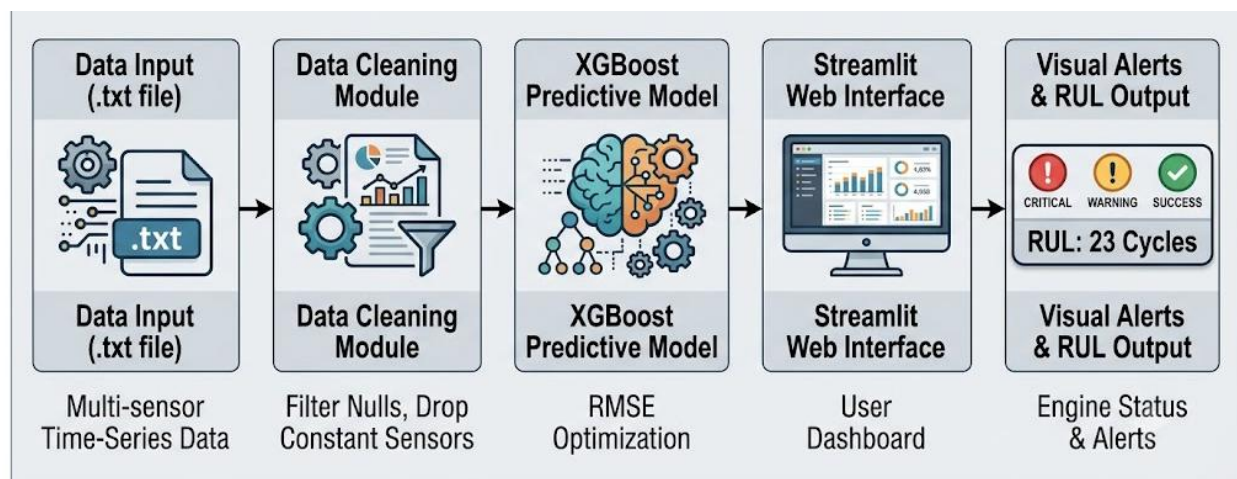


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

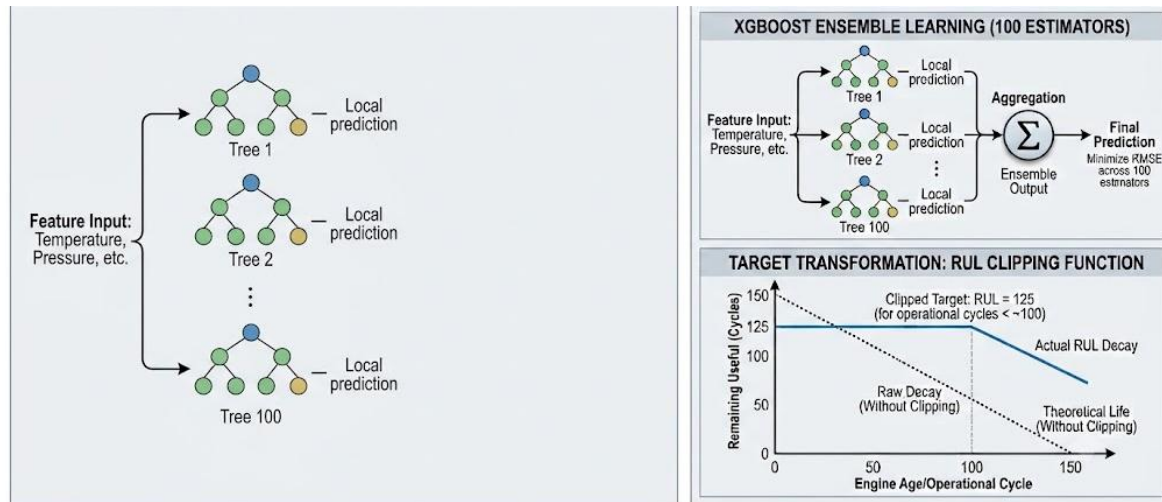


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

- Input Interface: A file upload module in the Streamlit sidebar accepting test_FD001.txt.
- Output Interface: A dynamic dashboard displaying the predicted RUL as a metric, accompanied by color-coded warning banners (Critical/Warning/Healthy) and a time-series plot of Sensor 14 degradation.

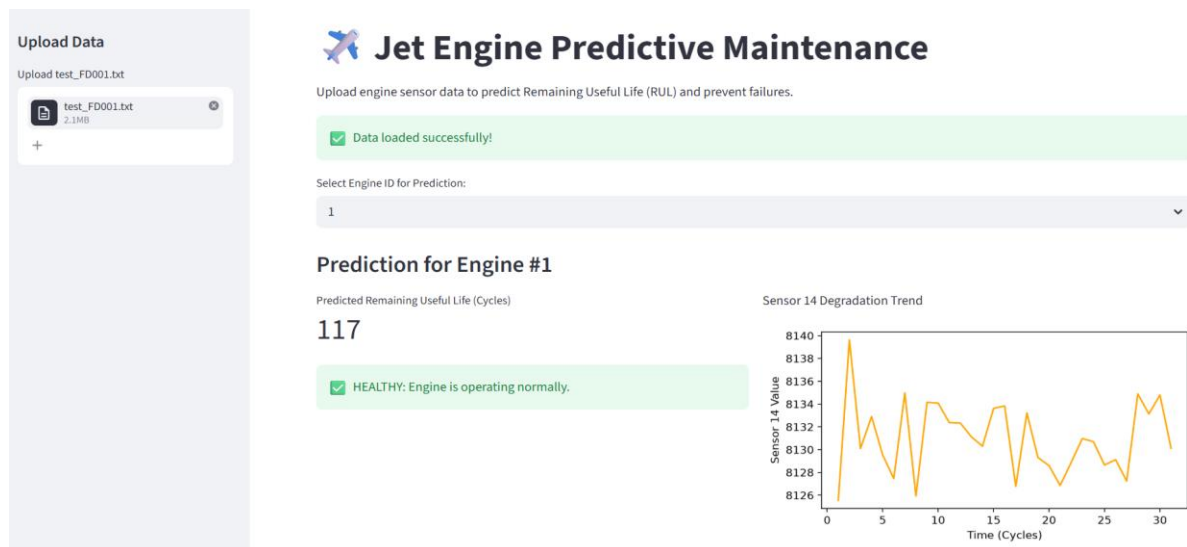


Figure 3: INTERFACES OF THE SYSTEM

6 Performance Test

To make this an industry-ready solution, handling constraints was critical. The primary constraint identified was the "Curse of Dimensionality" and model overfitting when too many engineered features (like rolling averages) were added. To take care of this, I restricted the model to the base sensors and optimized hyper-parameters ($\text{max_depth}=5$, $\text{n_estimators}=100$) to maintain speed and accuracy.

6.1 Test Plan/ Test Cases

- Test Case 1 (Healthy Engine): Input data of an engine with >60 cycles left. Expected result: "HEALTHY" status.
- Test Case 2 (Failing Engine): Input data of an engine with <30 cycles left (e.g., Engine #24). Expected result: "CRITICAL" red alert and downward/upward spike in sensor plots.

6.2 Test Procedure

The test_FD001.txt dataset was uploaded into the live Streamlit application. Various engine IDs were selected from the dropdown menu to simulate real-time monitoring, and the predicted RUL and alerts were observed on the dashboard.

6.3 Performance Outcome

The baseline model successfully achieved an RMSE (Root Mean Square Error) of 18.10 cycles. The Streamlit interface processed the data and rendered predictions instantly without any memory crashes, passing all UI test cases.

7 My learnings

This project was a massive learning curve. Key takeaways include:

- **Data Engineering:** Understanding how to handle complex time-series data and the importance of clipping target variables (RUL) to represent physical realities.
- **Algorithm Selection:** Learning why XGBoost outperforms standard Random Forests on tabular data.
- **Software Deployment:** Learning how to tackle real-world deployment bugs (like corrupted input streams and OS path issues) and successfully building a frontend using Streamlit.

8 Future work scope

Due to time constraints, the current deployed app handles the FD001 dataset (single operating condition). In the future, I plan to:

1. Implement Deep Learning (LSTM networks) to handle the complex FD004 dataset, which contains 6 operating conditions and multiple fault modes.
2. Deploy the Streamlit application on a cloud platform (like AWS or Render) so that it can be accessed globally via a public URL.